BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Midterm
Duration: 80 minutes

Set - A

Semester: Summer 2024
Full Marks: 40

CSE 420: Compiler Design

Figures in the right margin indicate marks.

Answer all the questions

COs	Questions	Marks
CO2	1. Construct the DFA from the following RE using **Direct Method** (a | ε) (a | b | c) (b | c)*	15
CO3	2. Consider the following *SLR Grammar*. Draw **LR(0) automaton** and construct **SLR parse table** 1. $T \rightarrow B\ C$ 2. $B \rightarrow$ ***int*** 3. $B \rightarrow$ ***float*** 4. $C \rightarrow \varepsilon$ 5. $C \rightarrow$ [**num]** C	10
CO3	3. Consider the **SLR parse table** from **Question 2**. Show the parsing simulation using **stack** for the input string, ***int [num][num][num]***	10
CO1	4. Explain the interaction between **Lexical analyzer** and **Parser**	5

Augmentation:

$T' \to T$

$T \to B\,C$

$B \to int$

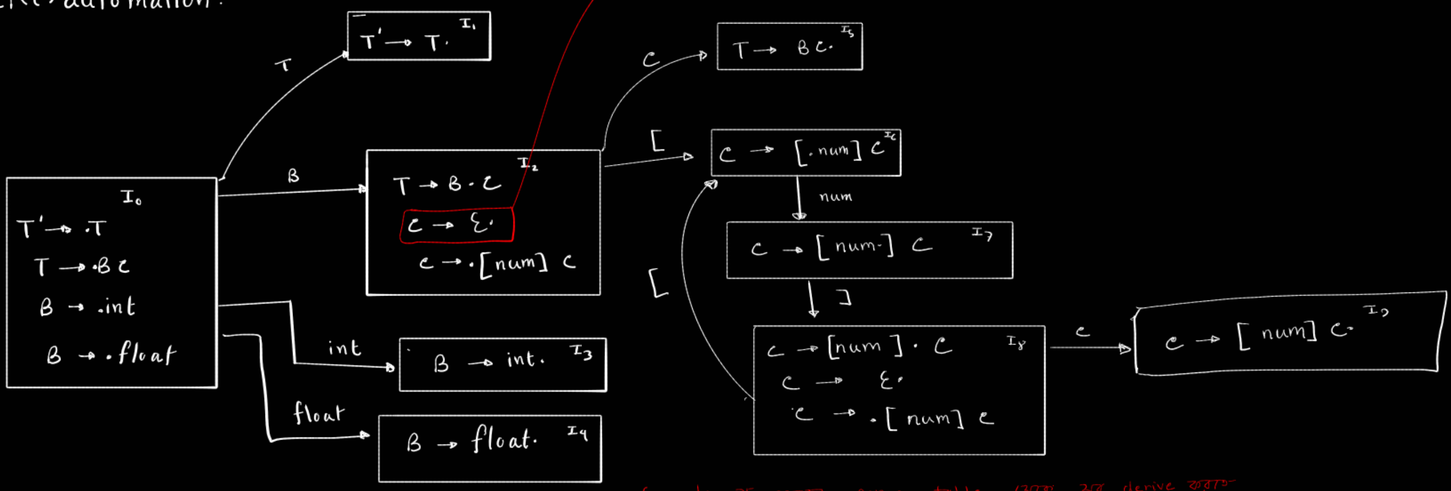
$B \to float$

$C \to \varepsilon$

$C \to [num]\,C$

এটা আর এটা same কথা; so direct reduce at this state.

LR(0) automation:



ε rule না থাকলে parse table থেকে সব derive করতে পারতাম। বা এখানে ε rule আছে so আমাদেরকে normally reduce করা লাগতে পারে।

First (T) = { int, float }

First (B) = { int, float }

First (C) = { [, ε }

Follow (T) = { $ }

Follow (B) = { [, $ }

Follow (C) = { $ }

$T \to B\,C$ (1)

$B \to int$ (2)

$B \to float$ (3)

$C \to \varepsilon$ (4)

$C \to [num]\,C$ (5)

state	Action						Go - To		
	[	num	]	int	float	$	T	B	C
0				S3	S4		1	2	
1						Accept			
2	S6					R4			5
3	R2					R2			
4	R3					R3			
5						R1			
6		S7							
7			S8						
8	S6					R4			9
9						R5			

Stack	Symbol	Input buffer	Action
0	$	int [num] [num] [num] $	S3
0 3	$ int	[num] [num] [num] $	Reduce by B → int
0 2	$ B	[num] [num] [num] $	S6
0 2 6	$ B [	num] [num] [num] $	S7
0 2 6 7	$ B [num	] [num] [num] $	S8
0 2 6 7 8	$ B [num]	[num] [num] $	S6
0 2 6 7 8 6	$ B [num] [	num] [num] $	S7
0 2 6 7 8 6 7	$ B [num] [num	] [num] $	S8
0 2 6 7 8 6 7 8	$ B [num] [num]	[num] $	S6
0 2 6 7 8 6 7 8 6	$ B [num] [num] [	num] $	S7
0 2 6 7 8 6 7 8 6 7	$ B [num] [num] [num	] $	S8
0 2 6 7 8 6 7 8 6 7 8	$ B [num] [num] [num]	$	Reduce by C → ε
0 2 6 7 8 6 7 8 6 7 8 9	$ B [num] [num] [num] C	$	Reduce by C → num [C]
0 2 6 7 8 6 7 8 9	$ B [num] [num] C	$	Reduce by C → num [C]
0 2 6 7 8 9	$ B [num] C	$	Reduce by C → num [C]
0 2 5	$ B C	$	Reduce by T → B C
0 1	$ T	$	Accept